Phạm Thanh Lâm

Network Engineer · Computer Networks & Data Communications

UIT graduate with hands-on experience in network security, high-availability infrastructure, and SDN. Seeking a full-time role in network engineering or security operations.

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Dĩ An, Bình Dương

EXPERIENCE

FSI — Công ty Cổ phần Công nghệ FSI
Mar 2026 - Jun 2026

Network & Security Engineer Intern

26/22 Đường số 03, Cư Xá Đô Thành, Phường Bàn Cờ, TP. HCM

- Configuring and maintaining enterprise-grade network infrastructure in a production environment.
- Hands-on work with Cisco switches, SD-WAN solutions, and security appliances under senior engineer guidance.
- Supporting deployment and troubleshooting tasks across client sites, including data center operations.

PROJECTS

HA Firewall & AI-driven Security Monitoring — Thesis (KLTN)
Jan - May 2026

- Built **Active-Passive OPNsense cluster** (CARP/pfsync) with HAProxy SSL offloading and **ModSecurity WAF (OWASP CRS)** for defense-in-depth; measured failover time and pfsync overhead.
- Deployed **Wazuh SIEM** pipeline; generated custom attack dataset (DVWA + Kali Linux); trained **Random Forest / Logistic Regression** classifier targeting FP rate reduction from >30% → <15%.

OPNsense · CARP/pfsync · HAProxy · ModSecurity · OWASP CRS · Wazuh SIEM · Random Forest · Python · VMware

SDN Server Load Balancer — Programmable Networks Course
2025

- Implemented L3 load balancer using **POX OpenFlow Controller + Mininet** (OVS); controller handles ARP/ICMP, selects backend via bidirectional NAT, pushes flow rules to switches.
- Benchmarked scheduling: **Round Robin** — 3-packet variance across 6 servers (82–84 tx_packets); **Random** — up to 60-packet imbalance (~50% skew). Health Check auto-reroutes on link-down with zero interruption.
- Wrote a **Python/Matplotlib script** parsing JSON logs exported from POX/Mininet to visualize per-server load distribution.

Python · POX/OpenFlow · Mininet · Open vSwitch · NAT · Matplotlib

Network Monitoring — LibreNMS
Sep - Nov 2024

- Deployed centralized NMS on Linux/Docker; configured **SNMP v2c/v3** for bandwidth, CPU, RAM collection; built Email/Telegram alerting and optimized incident response workflow.

LibreNMS · Docker · SNMP v3 · Linux

Enterprise Infrastructure — FSI
2026

- IOS upgrades (Install Mode) + **StackWise** on Cisco Catalyst 9200L in live production; **Peplink SpeedFusion** Bandwidth Bonding & Hot Failover for multi-branch WAN resilience.
- Deployed **F-DDH BOX** for C2 communication blocking; fiber optic troubleshooting at Viettel IDC.

Cisco IOS · StackWise · Peplink SD-WAN · F-DDH BOX · Data Center

CERTIFICATIONS & LEARNING

Networking Basics Cisco Networking Academy	In progress
Networking Devices and Initial Configuration Cisco Networking Academy	In progress
Endpoint Security Cisco Networking Academy	In progress
Network Defense Cisco Networking Academy	In progress

EDUCATION

University of Information Technology (UIT – VNU-HCM)

B.Sc. Computer Networks & Data Communications

GPA 7.72 / 10 · 2022 - 2026

Relevant courses:
Network Performance Evaluation · DevOps · Programmable Networks · Systems Programming

SKILLS

NETWORKING

TCP/IP · Routing/Switching · High Availability · CARP/pfsync · SDN/OpenFlow · SD-WAN · Cisco IOS · StackWise

SECURITY

OPNsense · Wazuh SIEM · ModSecurity WAF · HAProxy · Suricata IDS/IPS · LibreNMS · SNMP v2c/v3

SYSTEMS

Linux · VMware · Docker · Windows Server

DEV & TOOLS

Python · Shell Scripting · Git · Mininet · Matplotlib · scikit-learn

LANGUAGES

Vietnamese	Native
English	Technical reading